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// Arduino IR Sensor Code

#include <LiquidCrystal_I2C.h>
int IRSensor = 9; // connect ir sensor module  to Arduino pin 9
int buzzerPin=3;
int count=0;

LiquidCrystal_I2C lcd(0x27,16,2);
//Parameter Setup
void setup()
{
  pinMode(IRSensor, INPUT); // IR Sensor 1 pin INPUT
  lcd.init();
  lcd.begin(16,2);
  lcd.clear();
  lcd.backlight();      // Make sure backlight is on
   lcd.print("Students : ");
   pinMode(buzzerPin,OUTPUT);
}
void loop()
{
```

Fig. 3: Code snippet setting the pins and I2C address

```
//Parameter Setup
void setup()
{
  pinMode(IRSensor, INPUT); // IR Sensor 1 pin INPUT
  lcd.init();
  lcd.begin(16,2);
  lcd.clear();
  lcd.backlight();      // Make sure backlight is on
   lcd.print("Students : ");
   pinMode(buzzerPin,OUTPUT);
}
void loop()
{
  int sensorStatus = digitalRead(IRSensor); // Read the Sensor Status
  if (sensorStatus == 1 ) // Check Sensor is high or not
  {
    count++;
     lcd.setCursor(11,0);    //Set cursor to character 2 on line 0
     lcd.print(count);
     delay(1000);
       if (count > 40)
  {
    digitalWrite(buzzerPin,HIGH);
    delay(1000);
    digitalWrite(buzzerPin,LOW);
   lcd.clear();
```

Fig. 4: Code snippet setup and loop function

circuit diagram, as illustrated in Fig. 2. The circuit can be wired on a breadboard, utilising the power of Arduino Nano. Its key components include an Arduino Nano (MOD1), an LCD I2C RG1602A (MOD2), an IR sensor module (S1), and a piezo buzzer (B1). The piezo buzzer sounds the alert whenever the student count surpasses 40. Meanwhile, the LCD I2C visually displays the student count and conveys an overload message when the limit is breached.

Software

The programming for this counter is executed within the Arduino IDE. To begin, you must install the LCD library using the library manager. In the code, after including the LCD library, you'll need to define the IR pin number for input (usually set to 9). Next, specify the buzzer pin (usually pin 3) as an output. Also, set the I2C address for the LCD driver and determine the pixel size (typically 16x2). Refer to Fig. 3 for a snippet of the code, which sets the pins and I2C address.

In the code, the IR sensor detects the signal each time a student boards the bus, incrementing the count and displaying it on the LCD. When the student count reaches the predefined limit, an alert is triggered through the buzzer, indicating an overload situation. Refer to Fig. 4 for details on the code snippet, outlining the setup and loop function.

Construction and testing

After uploading the source code STUDENT_counter.ino to the Arduino Nano, assemble the counter on a breadboard or a general-purpose PCB. Once the circuit is correctly wired, enclose it in suitable housing, including the Arduino Nano board. Mount the LCD on the cabinet's front side for easy visibility. Power the circuit with a 5V supply from a laptop or desktop via a USB cable.

Monitor the student count on the LCD, which promptly activates an alarm when the count exceeds 40. This alert promptly notifies the conductor and the driver, prompting them to take necessary actions. The circuit was initially built on a breadboard, using jump wires, and has undergone extensive testing, exhibiting reliable performance across various scenarios. EFY

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

PARTS LIST

Semiconductors:		
MOD1	-	Arduino Nano board
MOD2	-	LCD I2C (RG1602A)
LED1	-	5mm LED
Miscellaneous:		
PZ1	-	Piezo buzzer
	-	Breadboard
	-	Jumpers wires
	-	IR sensor module
	-	5V USB power adaptor

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